

Optimizing Sales Forecasting and Inventory Management for a Small Pizza shop

A Proposal report for the BDM capstone Project

Submitted by

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Declaration Statement

I am working on a Project Title “**Optimizing Sales Forecasting and Inventory Management for a Small Pizza Shop**”. I extend my appreciation to the owner of **Mr. Pizza**, for providing the necessary resources that enabled me to conduct my project.

I hereby assert that the data presented and assessed in this project report is genuine and precise to the utmost extent of my knowledge and capabilities. The data has been gathered through primary sources and carefully analyzed to assure its reliability.

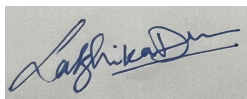
Additionally, I affirm that all procedures employed for the purpose of data collection and analysis have been duly explained in this report. The outcomes and inferences derived from the data are an accurate depiction of the findings acquired through thorough analytical procedures.

I am dedicated to adhering to the information of academic honesty and integrity, and I am receptive to any additional examination or validation of the data contained in this project report.

I understand that the execution of this project is intended for individual completion and is not to be undertaken collectively. I thus affirm that I am not engaged in any form of collaboration with other individuals, and that all the work undertaken has been solely conducted by me. In the event that plagiarism is detected in the report at any stage of the project's completion, I am fully aware and prepared to accept disciplinary measures imposed by the relevant authority.

I agree that all the recommendations are business-specific and limited to this project exclusively, and cannot be utilized for any other purpose with an IIT Madras tag. I understand that IIT Madras does not endorse this.

Signature of Candidate:



Name: Lakshika Dev

Date: 13-11-2025

2. Executive Summary

The organization is a small, independent pizza shop located in Kathgodam, Uttarakhand, managed solely by the owner. Established in 2023, it operates as a B2C food retail business offering a variety of pizza options and cold beverages through walk-ins and online platforms such as Zomato. Catering services for small events are also provided.

A significant problem has been identified due to manual and limited record-keeping, this is one of the main reasons for no effective sales forecasting, inventory management, and customer behavior analysis. This lack of data-driven decision-making has limited the operational efficiency, profitability, and growth potential.

To address these issues, a data-driven approach will be implemented by analyzing historical sales and purchase data. Techniques including time series analysis for sales forecasting, inventory turnover evaluation, and customer order pattern analysis will be done. Microsoft Excel and Python tools will be utilized for data cleaning, statistical analysis, and visualization. The insights derived are expected to optimize inventory purchasing, enhance marketing strategies, and improve operational efficiency. The project aims to help the owner with actionable recommendations for sustainable business growth, reduce waste, buy the right amount of ingredients, avoid stockouts and improve overall profit and daily operations.

3. Organization Background

The organization is a small local, independently owned pizza outlet located in Kathgodam, Uttarakhand established in 2023. The business is managed solely by the owner Mr. Prakash Singh Danu who handles production, packaging, and order-taking. The outlet functions as a B2C food-retail operation offering multiple pizza variants and cold beverages. Sales channels include direct walk-ins and online orders via Zomato. In addition to retail sales, the business provides catering services for small events and weddings, handling bulk orders occasionally.

Though newly established and small-scale, the shop follows food safety regulations and has certifications for the same. Despite challenges related to single-person management, the shop has secured a loyal customer base and maintains an active presence on popular delivery platforms. The owner is committed to business expansion and aims to utilize his cooking skills attained through Hotel Management degree to enhance product offerings and operational efficiency.

4. Problem Statement

4.1 Enhance sales forecasting accuracy by analyzing existing sales patterns and trends.

(The shop faces stockouts because it cannot predict daily or weekly demands properly.)

4.2 Optimize inventory management through evaluation of purchase and usage data to reduce wastage and costs.

(The shop needs to reduce ingredient wastage by understanding how much stock is actually required specially the stock related to dairy products.)

4.3 Improve customer retention strategies by looking at order trends and delivery channel effectiveness.

5. Background of the Problem

The pizza shop business currently operates with limited and manually maintained records, resulting in data gaps that hinder effective decision-making. The absence of systematic data collection restricts accurate sales forecasting and weakens inventory management, leading to potential inefficiencies such as stockouts or excess inventory. The owner works alone, handling everything from making pizzas to packaging, which makes it even harder to grow and avoid mistakes or delays.

External factors also contribute to the complexity of operations. The competitive food retail market, heightened by digital platforms like Zomato, demands fast response rates and high service quality, placing further pressure on operational efficiency. Ingredient costs fluctuate, making it difficult to control profit margins.

Addressing these issues through structured data analysis and operational optimization is important for improving resource utilisation, minimising waste, and enhancing customer satisfaction. Quantifiable insights from existing sales and purchase data will provide the foundation for data-driven strategies aimed at overcoming these internal and external challenges, therefore supporting sustainable business growth.

6. Problem Solving Approach

6.1 Methods:

A data-based plan will be used to fix these challenges. Sales forecasts will be made using past data to spot trends and busy seasons. Inventory will be managed by checking how purchases and usage relate to costs and waste. Customer orders will be studied to understand their preferences. Data from sales notes, purchase lists, and orders will be gathered, focusing on daily sales, item sales, ingredient purchases, and sales channel details.

Excel will be used to clean and organize the data, and Python tools will help analyze and create graphs. This will help turn raw data into clear advice for ordering stock, marketing, and daily work improvements. The plan is flexible and can be updated as new data comes in, helping the owner make better choices for better profits and customer satisfaction.

6.2 Data Collection:

Primary data will be sourced from sales notebooks, purchase records, and order logs available from the pizza shop. Key variables include daily sales volume, item-wise sales frequency, raw material purchase quantities (dairy items, toppings, etc), and sales channel performance (walk-in vs. Zomato orders). Additional qualitative data on bulk event sales and seasonal demand fluctuations will be collected through interviews with the owner. These data points are important because of their direct relationship to sales forecasting accuracy, inventory control, and customer satisfaction.

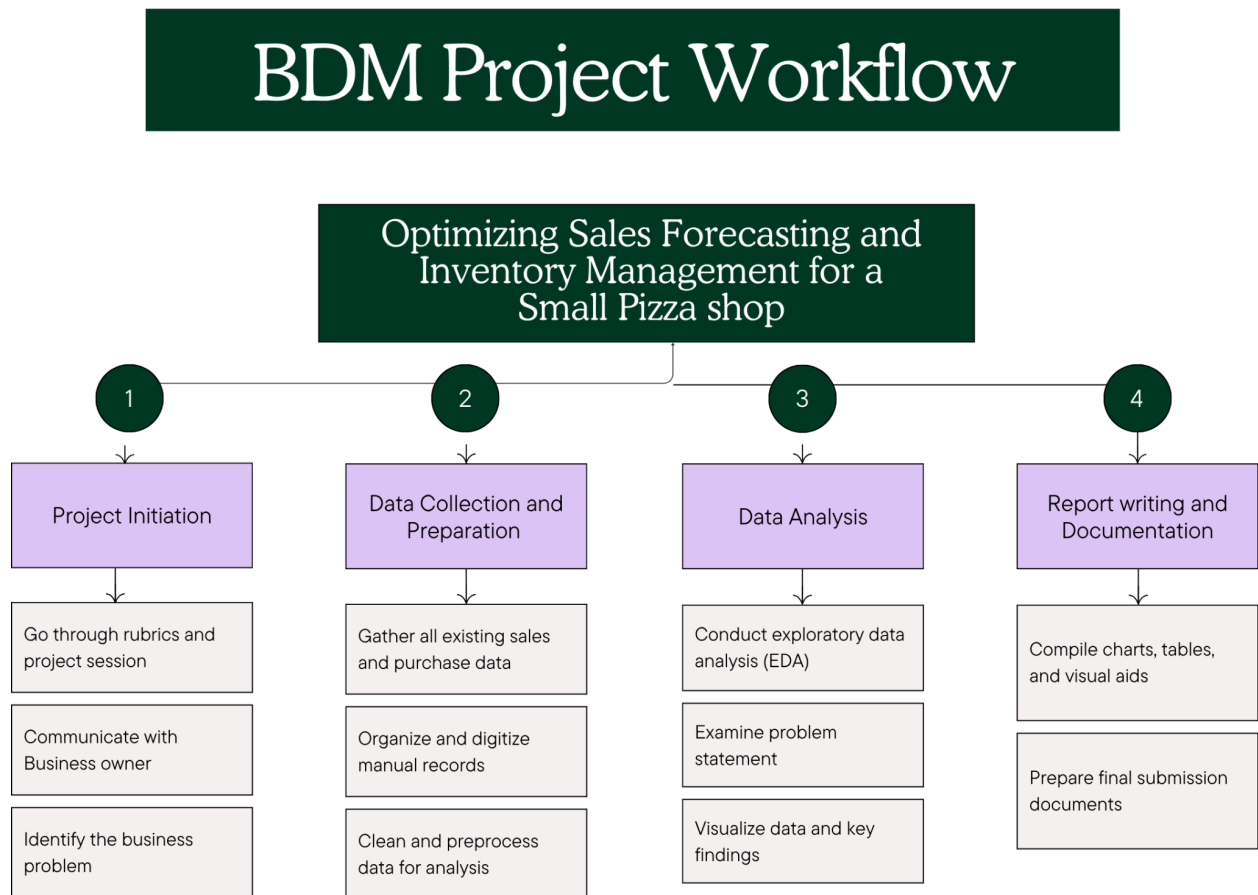
6.3 Analysis Tools:

Microsoft Excel will be used to clean and organize the data. Python, with libraries like Pandas, Matplotlib, Seaborn, and statsmodels, will be used to do the heavy statistical analysis and create graphs. Excel's features will help in summarizing and exploring the data, while Python will be used for advanced forecasting and spotting patterns. These tools are chosen because they are effective and easy to use for small business data.

This approach will transform raw data into clear insights that can improve stock ordering, marketing strategies, and daily operations. It also offers flexibility to update the plan as new data comes in to help the shop grow and make better decisions over time.

7. Expected Timeline

7.1 Work Breakdown Structure



7.2 Gantt Chart



8. Expected Outcome

This project aims to provide useful insights that will improve how the pizza shop runs its daily business. First, it is expected that better inventory planning will be possible by identifying which pizzas and ingredients are sold most often. This will help in buying the right amount of stock and cut down on waste, saving money in the long run.

Second, the project will highlight important sales trends and seasonal changes in customer orders. With this, the shop owner can plan production and staffing more effectively to meet demand without overproducing. Sales forecasting will provide a clearer picture of busy and slow days.

Third, by analyzing how different sales channels like in-shop walk-ins and online orders compare, the owner can decide where to focus marketing and service efforts.

Lastly, the project will give recommendations on how to handle bulk event catering better and maintain customer satisfaction. Overall, the data-driven recommendations will support the owner in making evidence-backed decisions to increase profitability, streamline operations, and plan for sustainable business growth, moving beyond reliance on intuition or informal records.